DHS/MIS malaria prevalence and child mortality analysis

Summary of main results

- **Higher malaria prevalence is associated with higher post-neonatal mortality.** In the ridge-linear model, a 10 percentage-point increase in PfPR2-10 is associated with 9.0% higher post-neonatal mortality (95% CI: 6.0%–12.2%). The estimate for neonatal mortality, our negative-control outcome, is small and statistically consistent with no association. See association results.
- **A model that allows the prevalence-mortality relationship to change over time fits best, but the evidence for this interaction is not decisive.** The selected spline model has lower AIC than the model without an interaction (ΔAIC 3.30) and a full `te(PfPR, year)` surface (ΔAIC 5.91); the nonlinear interaction has `p=0.057`. See model comparison.
- **Allowing a time interaction has little effect on the estimated cross-sectional relationship, but materially changes the estimated decline in malaria-attributable mortality.** From 2000 to 2024, estimated deaths fall by 52% with no time interaction, 20% with the selected `ti` interaction and 34% with the full `te` surface. See time-interaction results.
- **Our preferred model estimates about 611,000 malaria-attributable post-neonatal deaths for the latest period, but this estimate is uncertain and is not directly comparable to IHME or WHO.** The model-parameter 95% interval is 441,000–749,000. The point estimate is 30% higher than the IHME/GBD 2025 estimate and 41% higher than an approximate WHO under-five estimate; the interval overlaps both. The calculation evaluates the 2025 model surface using 2024 prevalence, mortality and birth inputs. See aggregate burden comparison.
- **Nigeria and DR Congo account for most of the aggregate difference from IHME and the WHO proxy, while the model gives lower estimates for several other countries.** This heterogeneity cautions against interpreting the aggregate difference as a uniform adjustment to existing estimates. See country-level comparison.
- **Our current bottom line is that the cross-sectional association is fairly robust, while the latest burden and especially the change since 2000 are less certain.** Alternative samples and covariate structures produce similar attributable fractions, but the time specification and alternative likelihood affect the temporal interpretation and precision. See sensitivity analyses and interpretation and limitations.

The sections below describe the data and methods underlying these claims, followed by the detailed results and sensitivity analyses.

Analysis structure

Data selection

The unit of analysis is a DHS/MIS survey-region-year.

1. Eligible DHS and MIS Births Recodes are surveys conducted from 2000 through 2025 in sub-Saharan Africa. AIS surveys are not included.
2. Survey regions are matched to DHS first-level survey boundaries.
3. Each region receives the population-weighted MAP PfPR2-10 estimate from the survey's calendar year. A missing 2025 MAP surface is flagged; 2024 is not silently substituted.

4. Regional all-under-five mortality and neonatal mortality are estimated from birth histories using the DHS synthetic-cohort life-table method. Post-neonatal mortality is calculated as U5MR minus NNMR. The birth-exposure denominator is retained for the count model offset.
5. The assembled validation panel contains 936 survey-region-years from 106 surveys and 35 countries. The shared primary sample requires:
   - country mean PfPR2-10 greater than 1%;
   - regional PfPR2-10 at least 1%;
   - positive finite birth exposure, U5MR, NNMR and post-neonatal mortality.
6. The resulting primary sample contains 921 region-years, 105 surveys and 34 countries, spanning 2000–2024.

Both the country-level and region-level prevalence flags remain in the analysis dataset. Sensitivity analyses separately include sub-1% regions and restrict the prevalence range to 5–40%.

Covariate processing

Twenty-one candidate covariates are evaluated before imputation. Variables with at most 5% missingness are retained and singly imputed using the within-country median, with the overall median as a fallback. Variables with more than 5% missingness are flagged and excluded.

The 16 included covariates are:

Level	Included covariates
Survey-region	Urban residence, regional DTP3 coverage, measles vaccination, facility delivery, maternal education, exclusive breastfeeding, birth interval under 24 months, maternal age at first birth, improved water, improved sanitation and household electricity
National, nearest year	DTP3 coverage, log GDP per capita, health expenditure as a share of GDP, log health expenditure per capita and electricity access

The five excluded variables are household wealth (5.56% missing), stunting (25.2%), underweight (27.5%), wasting (27.5%) and political stability (100%).

Proportions are logit transformed, continuous variables remain on their declared scale, and all included variables are standardised. They enter the model as one ridge-penalised matrix block; the amount of shrinkage is estimated from the data.

Main outcome model

For each survey region, the mortality rate is converted to an approximate death count using the DHS birth exposure. Models use:

- a negative-binomial likelihood with log link;
- `offset(log(exposure))`;
- a ridge penalty on the full eligible covariate block;
- country random intercepts;
- country-specific random linear PfPR slopes.

The four prespecified models and the full `te(PfPR, year)` sensitivity are compared using maximum likelihood and AIC on the same 921 observations:

Model	Prevalence and time specification	AIC	ΔAIC
Spline with `ti` interaction	`s(PfPR) + s(year) + ti(PfPR, year)`	7137.68	0.00
Linear with interaction	`PfPR + PfPR × year + s(year)`	7139.85	2.17
Spline without interaction	`s(PfPR) + s(year)`	7140.98	3.30
Full tensor surface	`te(PfPR, year)`	7143.59	5.91
Linear without interaction	`PfPR + s(year)`	7149.66	11.99

The full tensor surface incorporates the PfPR main effect, year main effect and their interaction in one term and uses 11.46 effective degrees of freedom. It is more flexible than the selected `s(PfPR) + s(year) + ti(PfPR, year)` decomposition but fits less well by AIC.

The selected specification is refitted with REML for post-neonatal mortality, all-under-five mortality and neonatal mortality. The linear no-interaction model is also refitted for all three outcomes to give an interpretable percentage change per 10 PfPR percentage points.

Main results

DHS/MIS association

From the ridge-linear summary models, each 10 percentage-point increase in PfPR2-10 is associated with:

Outcome	Change in mortality	95% CI	p-value
Post-neonatal	+9.04%	+5.98% to +12.19%	<0.001
All under-five	+6.63%	+4.27% to +9.04%	<0.001
Neonatal negative control	+0.95%	−1.19% to +3.14%	0.388

The neonatal result reduces, but does not eliminate, concern that the main association reflects broad differences in child survival between higher-prevalence and lower-prevalence settings.

Attributable fraction

The selected `ti` model estimates the following population-average fractions of post-neonatal mortality attributable to malaria relative to a 1% PfPR counterfactual:

PfPR2-10	2014 reference year	2025 surface
10%	14.6%	18.6%
30%	35.6%	43.0%
50%	43.0%	51.1%

Country random effects are set to zero for these population-average predictions and for the national burden estimates.

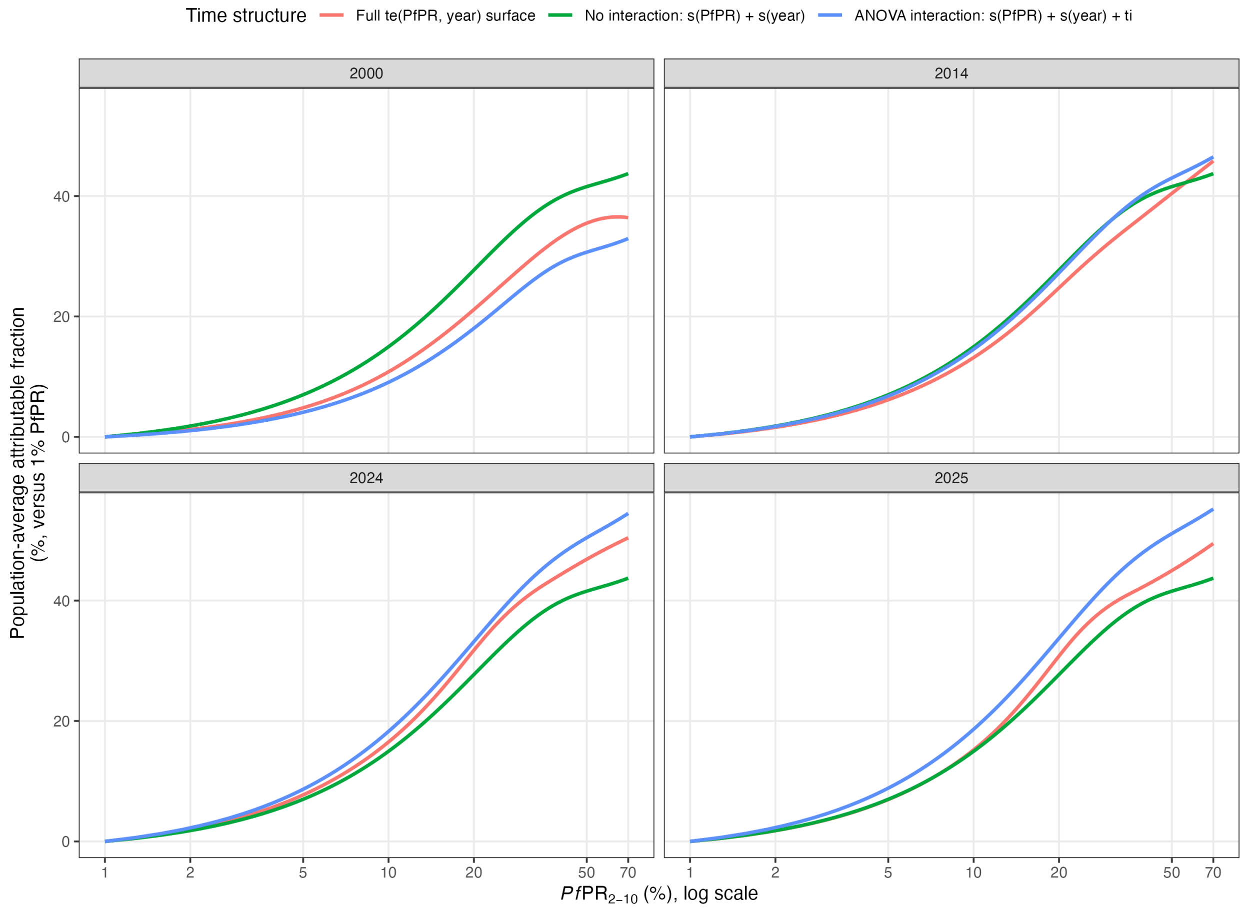

Figure 1: Attributable-fraction curves under the alternative time structures

Does the time interaction materially affect predictions?

Yes, particularly for historical trends and the latest aggregate burden, but the alternatives remain within overlapping uncertainty intervals.

At 30% PfPR, the attributable fraction is:

Model	2000	2014	2024	2025
No time interaction	35.7%	35.7%	35.7%	35.7%
Selected `ti` interaction	24.8%	35.6%	42.4%	43.0%
Full `te` surface	28.4%	32.2%	40.3%	39.1%

Applied to the same national prevalence and all-cause mortality series, the estimated post-neonatal malaria burden changes from 2000 to 2024 as follows:

Model	2000 deaths	2024 deaths	Change
No interaction	1,049,000	503,000	−52.1%
Selected `ti` interaction	753,000	601,000	−20.2%
Full `te` surface	867,000	570,000	−34.2%

This is the most important sensitivity for the manuscript's temporal bottom line. The claim of an approximately 50–57% decline follows the time-stable relationship. The AIC-selected time-varying relationship implies a much smaller decline of about 20%, while the full `te` surface lies between them.

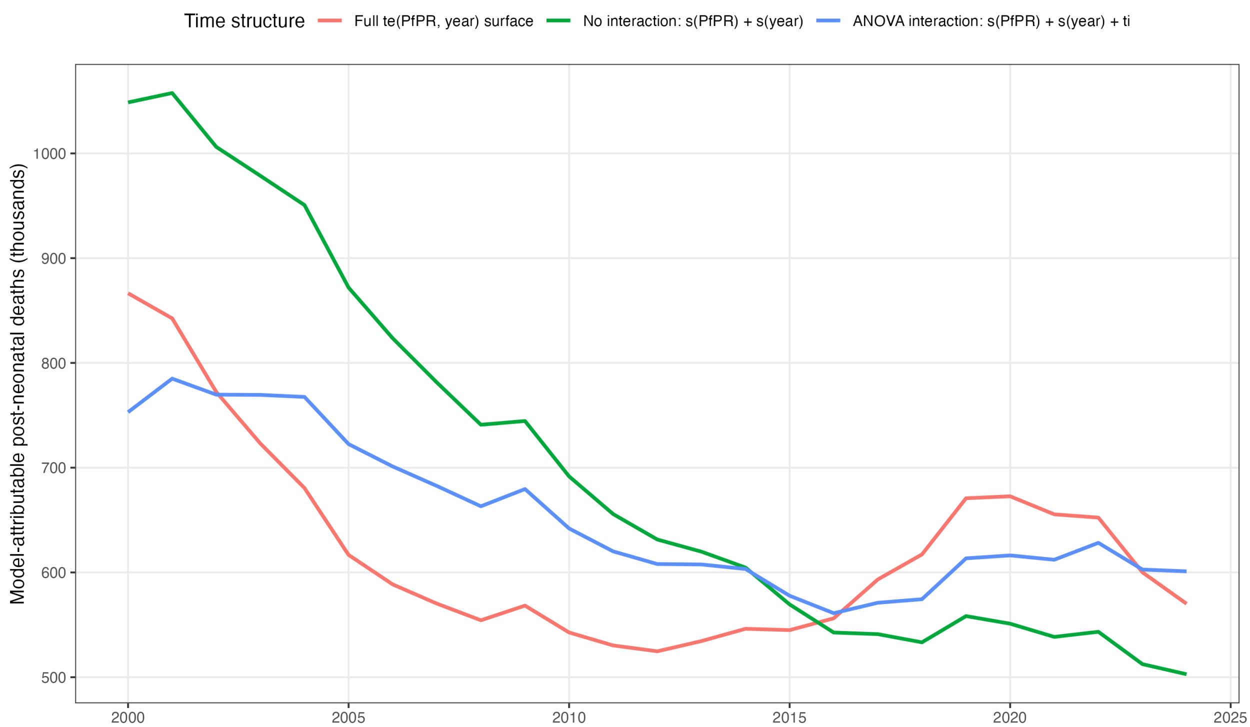

Figure 2: National burden trajectories under the alternative time structures

For the latest 2025-surface comparison, the estimates are:

Model	Estimated deaths	Model-parameter 95% interval
No interaction	503,000	370,000–621,000
Selected `ti` interaction	611,000	441,000–749,000
Full `te` surface	551,000	218,000–804,000

Thus the selected interaction raises the latest point estimate by about 108,000 deaths, or 21%, compared with no interaction. The full `te` estimate is about 60,000 lower than the selected `ti` estimate.

Other sensitivity analyses

Alternative samples

All sample sensitivities retain a positive nonlinear prevalence relationship:

- Primary sample: AF at 30% PfPR = 35.6%.
- Include regions below 1% PfPR: 33.8%.
- Restrict PfPR to 5–40%: 32.5%.
- Complete cases only: 35.6%.

These choices do not materially change the cross-sectional attributable fraction.

Model structure and covariate blocks

- Removing the country random PfPR slope worsens fit by about 16.7 AIC units.
- Regional covariates alone have the lowest AIC in the structural sensitivity set, 3.1 units below the full regional-plus-national block.
- National covariates alone fit very poorly (ΔAIC 219).
- Changing the PfPR spline basis from `k=6` to `k=4` or `k=8` changes AF at 30% only from about 34.3% to 35.1%.
- Across structural sensitivities, AF at 30% ranges from 33.2% to 42.8%; the upper value occurs in the national-covariate-only model, which fits poorly.

Alternative likelihood

The log-Gaussian rate sensitivity is less precise than the primary negative-binomial count model:

- ridge-linear post-neonatal estimate: +7.13% per 10 PfPR points (95% CI −0.60% to +15.47%; `p=0.072`);
- nonlinear prevalence term: `p=0.049`;
- its post-neonatal time interaction is not supported (`p=0.599`).

This weakens certainty about the exact effect size and the time interaction, but the point estimate remains positive.

Latest national burden comparison

Aggregate bottom line

The closest available comparison is not perfectly contemporaneous:

- **Model:** surface evaluated at 2025 using 2024 MAP PfPR, IGME mortality and live births for 43 matched sub-Saharan African countries.
- **IHME:** 2025 under-five malaria deaths.
- **WHO:** WMR 2025 estimates for 2024. WHO reports all-age country deaths, so the under-five comparator is an approximate 75% share.

Source	Deaths
Selected `ti` prevalence/all-cause model	**611,000**
IHME/GBD SSA under-five, 2025	470,000
WHO African-region all-age, 2024	579,000
WHO African-region under-five proxy, 2024	434,000

WHO's 2025 annex explicitly supplies country estimates through 2024, not 2025. WHO also states that children under five account for approximately 75% of malaria deaths in the African Region. Sources: WMR 2025 Annex 4H, WHO malaria fact sheet, and the WHO GHO `MALARIA_EST_DEATHS` country indicator.

Countries with the largest differences from IHME

The table below uses the AIC-selected `ti` model. "Very different" is defined in the generated results as at least a two-fold difference and an absolute difference of at least 1,000 deaths.

Country	Model	IHME 2025	Model/IHME	Interpretation
Nigeria	224,600	150,500	1.49	Largest absolute excess: +74,100

Country	Model	IHME 2025	Model/IHME	Interpretation
DR Congo	137,300	68,300	2.01	Model approximately twice IHME
Chad	15,900	5,500	2.89	Model substantially higher
Sudan	5,200	1,400	3.88	Model substantially higher
South Sudan	7,400	4,300	1.73	Model higher, below two-fold threshold
Uganda	17,000	28,500	0.60	Model lower by about 11,500
Burundi	4,900	15,700	0.31	Model substantially lower
Ethiopia	5,300	9,300	0.57	Model lower
Rwanda	400	2,500	0.15	Model substantially lower

Against the approximate WHO country-level under-five proxy, the clearest two-fold discrepancies are:

- higher under the model: DR Congo;
- lower under the model: Tanzania, Ethiopia, Rwanda, Madagascar and Kenya.

Nigeria and DR Congo together account for most of the model's aggregate excess over both IHME and the WHO proxy.

Interpretation and limitations

The comparison is not a validation against a common estimand:

- the prevalence/all-cause model is intended to include direct and indirect malaria-associated post-neonatal deaths;
- IHME and WHO estimate deaths assigned specifically to malaria;
- the model uses population-average effects and ignores country random effects for national extrapolation;
- 2025 exposure and all-cause mortality inputs are unavailable, and WHO's latest estimates are for 2024;
- the WHO country under-five comparison applies a regional 75% age-share assumption to every country.

Our current interpretation is that the data provide fairly robust evidence of a positive cross-sectional association between malaria prevalence and post-neonatal mortality. We place less weight on the exact contemporary total and, especially, the estimated decline since 2000 because these depend materially on how the prevalence-mortality relationship is allowed to change over calendar time.

Reproducible outputs

`R_dhs/10_time_surface_and_burden.R` fits the `te` sensitivity and produces the time-surface and national comparison tables.

Key result tables are stored as CSV files under `results/dhs_rebuild/`.